Sistema cardiovascular

Ecuaciones principales

$F_a(t) = F_z(t), F_L(t) = F_c(t) + F_R(t)$

$F_z(t) = \frac{P_a(t) - P_p(t)}{Z}$

$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$

$F_c(t) = C\frac{dP_p(t)}{dt}$

$F_R(t) = \frac{P_p(t)}{R}$

Transformada de Laplace

$F_a(s) = F_z(s) + F_L(s) = F_c(s) + F_R(s)$

$F_z(s) = \frac{P_a(s) - P_p(s)}{Z}$

$F_L(s) = \frac{P_a(s) - P_p(s)}{Ls}$

$F_c(s) = C_s P_s(s)$

$F_R(s) = \frac{P_p(s)}{R}$

Procedimiento algebraico

$F_z(s) - F_L(s) = F_c(s) + F_R(s)$

$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s)}{Ls} - \frac{P_p(s)}{Ls} = C_s P_p(s) + \frac{P_p(s)}{R}$

$P_a(s)\left(\frac{1}{Z} + \frac{1}{Ls}\right) = P_p(s)\left(\left(s + \frac{1}{R} + \frac{1}{Ls} + \frac{1}{Z}\right)\right)$

$P_a(s)\frac{Ls+Z}{LZs} = P_p(s)\frac{CLRZs^2 + LZs + RZ + RLs}{RLZs}$

$P_a(s)\frac{Ls+Z}{LZs} = P_p(s)\frac{CLRZs^2 + L(Z+R)s + RZ}{RLZs}$

$\frac{P_p(s)}{P_a(s)} = \frac{\frac{Ls+Z}{LZs}}{\frac{CLRs^2 + L(Z+R)s + RZ}{RLZs}}$

Resultado

$\frac{P_p(s)}{P_a(s)} = \frac{LRs + RZ}{CLRZs^2 + L(Z+R)s + RZ}$

Estabilidad en lazo abierto

$CLRZs^2 + L(Z+R)s + RZ \quad a = CLRZ,\ b = L(Z+R),\ c = RZ$

$\lambda_{1,2} = \frac{-L(Z+R) \pm \sqrt{L^2(Z+R)^2 - 4CLRZ^2}}{2CLRZ}$

Hipotenso

$z=0.010,\ L=0.001,\ R=1.000,\ C=3.500$

$\lambda_1 = -14.429 + 8.805i,\ \lambda_2 = -14.429 - 8.805i$

Respuesta estable y subamortiguada a un escalón unitario de entrada

Normotenso

$z=0.010,\ L=0.001,\ R=1.000,\ C=1.500$

$\lambda_1 = -12.062,\ \lambda_2 = -55.797$

Respuesta estable y sobreamortiguada a un escalón unitario de entrada

Hipertenso

$z=0.010,\ L=0.001,\ R=1.000,\ C=1.000$

$\lambda_1 = -11.177,\ \lambda_2 = -89.873$

Respuesta estable y sobreamortiguada a un escalón unitario de entrada

Error en estado estacionario

Se calcula a un escalón unitario de entrada

$$e(s)=\lim_{s\to 0} s\,Pa(s)\left(1-\frac{Pp(s)}{Pa(s)}\right)=e(s):\lim_{s\to 0} s\,\frac{1}{s}\left(1-\frac{LRs+RZ}{CLRs^2+L(Z+R)s+RZ}\right)=0$$

$e(t)=0V$

Modelo de ecuaciones integro-diferenciales

$$F_Z(t)+F_L(t)=F_C(t)+F_R(t)$$

$$\frac{Pa(t)}{Z}+\frac{1}{L}\int[Pa(t)-Pp(t)]dt-C\frac{dPp(t)}{dt}=\frac{Pp(t)}{R}+\frac{Pp(t)}{Z}$$

$$\left(\frac{1}{R}+\frac{1}{Z}\right)Pp(t)=\frac{Pa(t)}{Z}+\frac{1}{L}\int[Pa(t)-Pp(t)]dt-C\frac{dPp(t)}{dt}$$

$$Pp(t)=\left[\frac{Pa(t)}{Z}+\frac{1}{L}\int[Pa(t)-Pp(t)]dt-C\frac{dPp(t)}{dt}\right]\frac{RZ}{Z+R}$$